

often worth the extra processor time. Global Illumination is controlled in two places. First, the light must be set up to cast photons; then the renderer must be configured to accept these photons and render the results.

5.6.2 Photons

There are two main lighting parameters to consider when setting up global illumination: energy and number of photons. Energy is the brightness of the light, so more energy produces a brighter global illumination solution. The number of photons is similar to a sampling rate and affects the quality of the image. More photons produce more accurate results, but at the cost of additional render time. For physically accurate renders, the number of photons can get incredibly high (into the millions). For example, when the number of photons is low, individual spots of light can be seen. As the number of photons increase, they start to blend and eventually overlap, and the illumination can be seen. When using just photons to illuminate the scene, it can take millions of photons to completely smooth out the graininess.

5.6.3 Energy

The second major attribute used to calculate global illumination is energy. This is roughly equivalent to a light's brightness. Photons, however, perform only indirect lighting, so increasing energy will increase only the amount of indirect light in the scene. Getting the right illumination requires getting the correct balance between the direct light and indirect light.

Energy is sensitive to the scale of the scene. Larger scenes need much larger energies to get the same amount of lighting as a smaller scene. In addition to the scale of the scene, energy also works in conjunction with the decay value, which determines how quickly the energy dissipates in the scene. A decay of 2 creates an inverse square falloff, which simulates the way real light illuminates a scene. Lower values create more light; higher values attenuate the light.

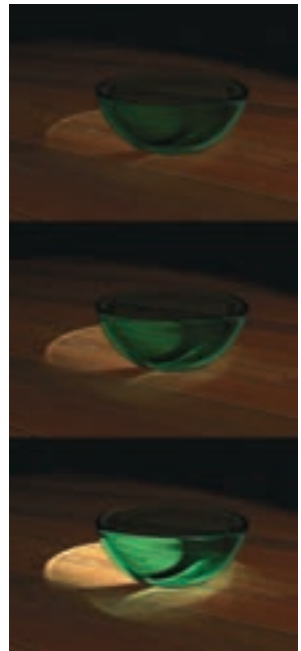


Image rendered with Decay = 2 and energy 4000, 8000 and 16000 respectively from top to bottom